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Yenepoya Institute Of Arts Science Commerce & Management

PROJECT – HIGH LEVEL DESIGN

ON

MULTI-CLOUD HYBRID STRATEGY FOR HEALTHCARE SYSTEM

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Project Code: PRJN26-83

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Table of Contents:

1.Introduction._____ 4-5
1.1. Scope of the Document.
1.2. Intended Audience.
1.3. System Overview.

2.System Design._____ 5-9

2.1 Application Design .
2.2 Process Flow .
2.3 Information Flow .
2.4 Components Design .
2.5 Key Design Considerations .
2.6 API Catalogue .

3.Data Design_____ 9-10

3.1 Data Model
3.2 Data Access Mechanism .
3.3 Data Retention Policies .
3.4 Data Migration .

4.Interfaces_____ 10-11

5.State and Session Management_____ 11-12

6.Caching_____ 12-13

7.Non-Functional Requirements_____ 13-14

7.1 Security Aspects
7.2 Performance Aspects

8. References_____ 14-15

9.System Architecture_____15-16

9.1.Architecture Overview
9.2.Architecture Flow

10.Deployment Architecture_____ 16

10.1.Deployment Components
10.2.Deployment Diagram Explanation
10.3.Deployment Benefits

1. Introduction

1.1 Scope of the Document

This document provides the **High-Level Design (HLD)** for the project titled **"Multi-Cloud Hybrid Backup Strategy for Healthcare System."**

It includes:

- System architecture
- Backup and restore mechanisms
- Redundancy and failover strategies
- Data handling and workflows

1.2 Intended Audience

- Students (BCA/BSc)
- Developers
- Industry Mentors / Evaluators
- Cloud & DevOps Engineers

1.3 System Overview

The system is a **Django-based Healthcare Management Application** integrated with a **Multi-Source Backup System**.

Features:

- Doctor & patient management
- Appointment booking
- Prescription upload
- Automatic backup trigger
- Local + Cloud backup

- Failover restore

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2. System Design

2.1 Application Design

Layered Architecture:

- **Presentation Layer** → UI (HTML templates)
- **Application Layer** → Django logic
- **Backup Service Layer** → Python scripts
- **Storage Layer** → Local + Cloud

2.2 Process Flow

Backup:

Doctor → Upload Prescription → Signal → Backup → Local + Cloud

Restore:

Local Backup → If Fail → Cloud Backup → Restore

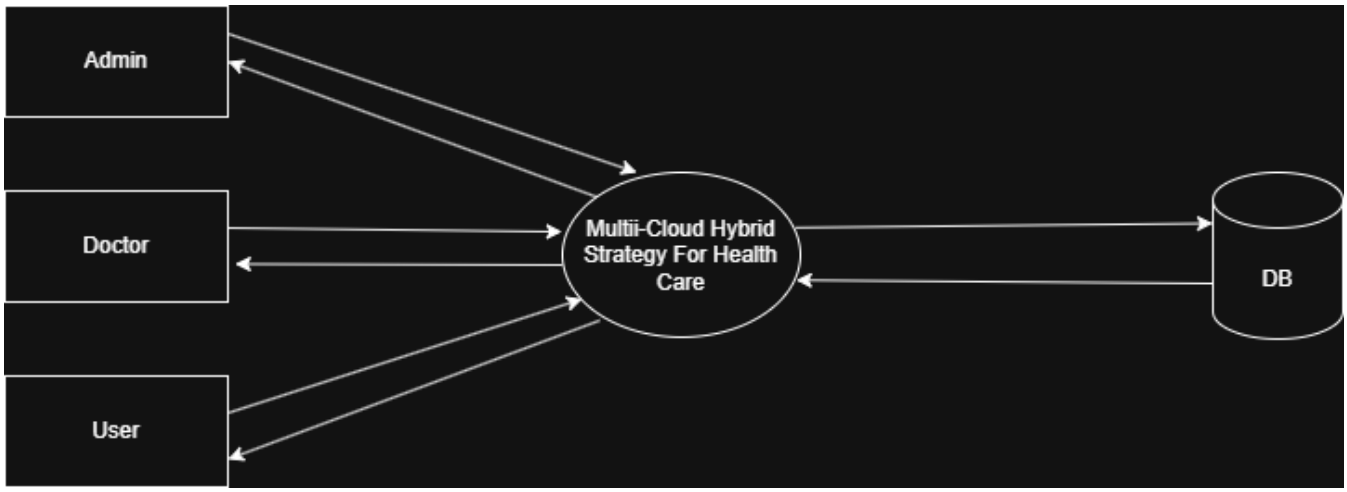
2.3 Information Flow

User → Django App → Database
→ Trigger Signal → Backup Module
→ Local Storage + Cloud Storage

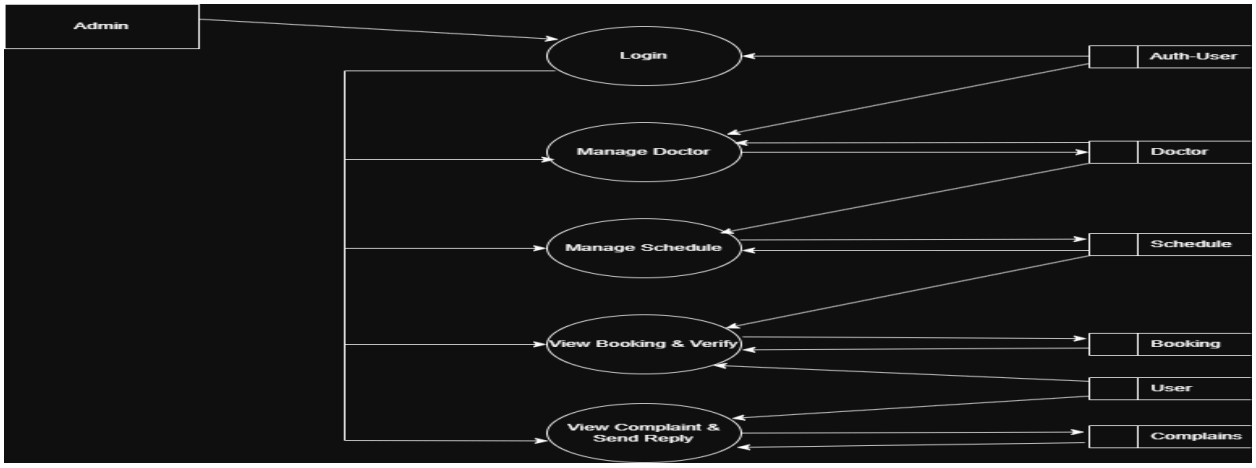
DATA FLOW

- Level 0
- Level 1
- Level 2
- Level 3

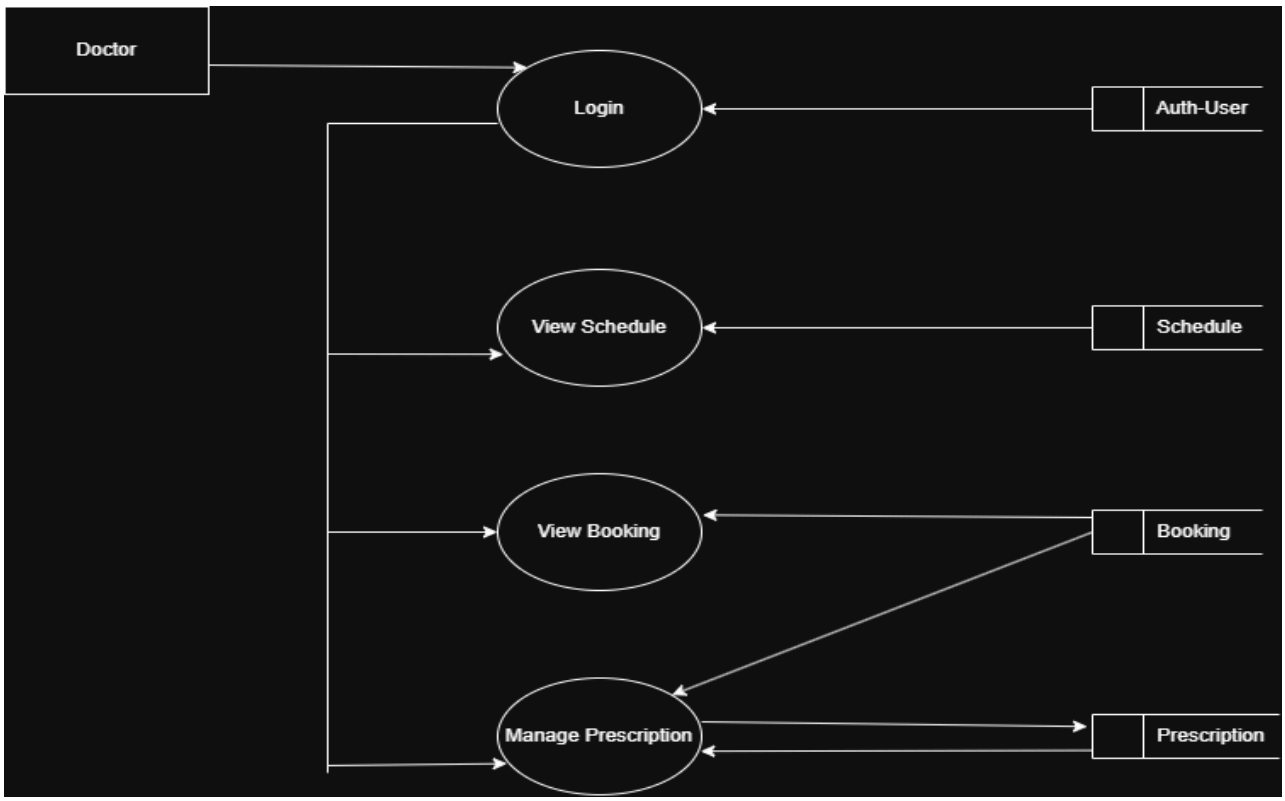
Level 0:-



Level 1:-



Level 2:-


Doctor
Login
Auth-User
View Schedule
Schedule
View Booking
Booking
Manage Prescription
Prescription

Level 3

2.4 Components Design

- Django Application
- Backup Module
- Restore Module
- Scheduler
- Cloud Integration

2.5 Key Design Considerations

- Reliability
- Redundancy
- Failover

- Scalability
- Security
- Performance

2.6 API Catalogue

- `run_full_backup()`
- `backup_database()`
- `backup_media()`
- `_upload_drive()`
- `run_restore()`
- `trigger_backup_async()`

---

3. Data Design

3.1 Data Model

Entities:

- Doctor
- User (Patient)
- Schedule
- Booking
- Prescription
- Complaint

Relationships:

- Doctor → Schedules
- User → Bookings
- Booking → Prescription

- User → Complaints

3.2 Data Access Mechanism

- CRUD using Django ORM
- Database backup using `mysqldump`
- Media handled via file system

3.3 Data Retention Policies

- Keep last **3 DB backups**
- Keep last **2 media backups**
- Old cloud backups deleted

3.4 Data Migration

- Backup → Store → Transfer → Restore
- Failover to cloud if local fails

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4. Interfaces

4.1 User Interface

- Admin
- Doctor
- Patient

4.2 System Interface

- Django ↔ Database (ORM)
- Django ↔ Backup Module (Signals)

4.3 External Interfaces

- Google Drive API
- MySQL Database

4.4 File System Interface

- Stores `.sql` and `.zip` backups locally

4.5 Backup & Restore Interface

- Automatic backup trigger
- Command-based restore

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5. State and Session Management

5.1 Session Management

- Django session framework
- Session ID via cookies
- Auto expiry and logout

5.2 Authentication

- Username/password login
- Role-based access:
  - Admin
  - Doctor
  - User

5.3 State Management

- Database state
- Backup state
- Application state

5.4 Backup Trigger State

- Triggered using `post_save` signal

5.5 Failover Handling

- Local backup priority
- Cloud fallback

---

6. Caching

6.1 Purpose

- Improve performance

- Reduce DB load

6.2 Types

- Application caching
- File system caching
- Session caching

6.3 Implementation

- Django caching (optional)
- Local backup as cache

6.4 Cache Expiry

- Updated when data changes
- Controlled via retention policy

6.5 Future Enhancements

- Redis / Memcached
- Distributed caching

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7. Non-Functional Requirements

7.1 Security Aspects

- Secure login authentication
- OAuth for cloud access
- Protected database configuration
- Role-based access control

7.2 Performance Aspects

- Asynchronous backup execution
- Reduced response time using caching
- Efficient file handling

7.3 Reliability

- Multi-source backup
- Failover mechanism

7.4 Scalability

- Supports cloud expansion (AWS S3, Azure)

7.5 Availability

- Backup ensures data availability
- Restore mechanism ensures recovery

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8. References

- Django Documentation
- Python Documentation
- Google Drive API Docs
- MySQL Documentation
- APScheduler Documentation

9. System Architecture (Extended)

The proposed healthcare backup system follows a **multi-layered architecture** integrating web application components with backup infrastructure. This design ensures scalability, reliability, and maintainability.

9.1 Architecture Overview

The system is divided into the following layers:

- **Presentation Layer**
  - Handles user interactions through web interfaces.
  - Built using HTML, CSS, and Bootstrap.
  - Used by Admin, Doctor, and User roles.
- **Application Layer**
  - Implements business logic using Django framework.
  - Handles authentication, booking, prescription, and scheduling logic.
- **Data Layer**
  - Stores structured data in MySQL database.
  - Includes tables such as doctor, booking, prescription, etc.
- **Backup Layer**
  - Responsible for data backup and recovery.
  - Stores backup in:
    - Local Storage
    - Cloud Storage (Google Drive)

9.2 Architecture Flow

1. User interacts with frontend interface.
2. Request is processed by Django views.
3. Data is stored/retrieved from database.
4. Important events trigger backup system.
5. Backup is stored locally and uploaded to cloud.

This layered design ensures **clear separation of responsibilities**.

10.Deployment Architecture

The system is designed to be deployed in a flexible environment suitable for academic and small-scale healthcare systems.

10.1 Deployment Components

- Web Application Server (Django)
- Database Server (MySQL)
- Local Storage System
- Cloud Storage (Google Drive API)

10.2 Deployment Diagram Explanation

- Django application runs on a local server or cloud VM.
- MySQL database runs either locally or remotely.
- Backup module interacts with:
    - Local filesystem
    - Google Drive via API

10.3 Deployment Benefits

- Easy setup for students (local environment)
- Scalable to cloud platforms
- Supports hybrid storage approach